

REPORT ON TESTING OF SMART PILLBOX

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Team Present: IN Google meet

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1. Introduction

The purpose of this testing session was to evaluate the functioning, connectivity, and reliability of the smart pillbox prototype. The device had been previously shipped to us, and this round of testing was intended to verify basic hardware operation, power system behavior, ESP microcontroller activity, and Bluetooth data communication.

This report documents the observations, issues identified, troubleshooting steps performed, and partial outcomes achieved during the session conducted on **3 December 2025**.

2. Initial Setup and Power-On Attempts

At the beginning of the test, the team faced difficulty in powering on the smart pillbox. The device did not respond on initial attempts to activate it. After examining the physical structure, the team applied pressure on the **yellow lid**, suspecting a loose power connection.

Upon pressing the lid and connecting the device to a **USB-C charging port**, the pillbox displayed a **blue activation indicator**, confirming that some portion of the power system was active. However, despite this LED response, the **ESP module did not show any signs of functioning**, indicating a deeper issue in internal circuitry or power delivery.

3. Inspection of Internal Components

To investigate the cause behind the ESP's non-responsiveness, the team proceeded to open the **blue plastic lid** of the pillbox.

3.1 Findings

- Inside the pillbox, the **battery's ground wire was found to be distorted or loose**, most likely due to mishandling or vibration during shipping.
- This faulty ground connection prevented stable power supply to the internal circuit.
- As a result, the ESP microcontroller remained inactive even when the external USB-C port showed a charging indicator.

Identifying this fault confirmed that the issue was hardware-related and not due to firmware malfunction.

4. ESP Functionality Testing

To further confirm whether the ESP module was operational, it was removed from the pillbox power line and tested independently.

4.1 Procedure

- The ESP board was connected directly to a computer using a **USB-B cable**.
- This ensured stable external power and allowed direct serial communication for testing.

4.2 Result

- The ESP powered on through the USB-B connection.
- This verified that the ESP board itself was not damaged.

Thus, the primary cause of non-functionality was confirmed to be the broken ground connection inside the pillbox.

5. Bluetooth Connectivity Tests

5.1 Testing via Android Teams App (Bluetooth Admin)

- The team attempted pairing and communication using the **Android Teams Bluetooth Admin app**.
- While the app could detect the device intermittently, it **failed to read or transmit any data**.
- This indicated incomplete or unstable Bluetooth communication.

5.2 Testing via Custom Smartphone App

- The pillbox was then tested using the **custom app developed for the project**.
- The custom app successfully **detected the Bluetooth module**, confirming that the device was broadcasting.
- However, **no data packets were received**, indicating either:
 - a serial communication issue between ESP and Bluetooth module, or
 - unstable power supply because of the internal wiring fault.

5.3 Testing via Third-Party App: Serial Bluetooth Terminal

- A third-party application named “**Serial Bluetooth**” from the Play Store was used.
- All available Bluetooth slots were scanned.
- The app initially **failed to detect stable internal BC signals**, again due to power instability.
- After multiple attempts, partial detections occurred, showing that the module was broadcasting but not fully functional.

These tests collectively verified that the Bluetooth module was functional but could not maintain stable communication due to internal wiring and power issues.

6. Conclusion

The testing session on **3 December 2025** revealed that the smart pillbox suffered from a **critical internal power connection failure**, specifically a **distorted or loose battery ground wire** caused during shipping. This resulted in:

- ESP microcontroller failing to boot through onboard power
- Unstable Bluetooth communication
- Intermittent device activation despite USB-C charging indicator

Independent testing confirmed that the ESP and Bluetooth module themselves are functional. The root cause of malfunction is therefore the damaged internal ground wiring.

Next Steps / Recommendations

1. **Repair the ground wire connection** and secure all internal wiring with protective insulation.
2. Add **shock-proof padding** inside the pillbox to prevent shipping-related damage.
3. Retest the ESP’s power-on, Bluetooth communication, and app data transmission after repair.
4. If available, update firmware to log power failures for easier diagnostics.
5. Perform a complete end-to-end test once the internal circuit is stable.

